

- A separate mandate for purchasing Hydropower **(HPO)**: now power utilities are obligated to **maintain a minimum % of hydro projects in their power purchase portfolio.**
- **Tariff rationalization:** Hydro projects typically last 60 to 100 years. Accordingly, the tariff can be rationalized by increasing the debt payment period.
- The policy provides flexibility to the developers to determine tariffs by back loading of tariff after increasing project life to 40 years, increasing the debt repayment period to 18 years, and introducing an escalating tariff of 2%
- **Budgetary support** is provided for **funding flood moderation component of hydropower projects** and for **funding cost of enabling infrastructure** i.e. roads & bridges on case to case basis & as per project capacity.
- A large number of projects which were languishing have been taken back from the existing developers and allotted to new developers.
- Hydro projects would get stuck because of geological surprises and resultant contracted disputes. To address this :-
  - We put in place the institution of **Independent Engineers** who resolve issues after hearing both parties and inspecting the site.
  - **A conciliation committee of Independent Experts** has been formed **for settlement of any dispute which is not resolved by the Independent Engineer**.
  - **We have notified guidelines** suggesting measures like preparation of realistic project schedule based on experience of projects in the vicinity, and use of latest Software tools like Primavera, MS-Project etc. for monitoring of projects regularly.
  - **Development of an IT-based monitoring system at the national level** to monitor the progress of under-construction HEPs.
  - **A system of regular site visits/ interaction by Ministerial & CEA officials** to monitor and facilitate faster resolution of inter-ministerial issues.
  - **Incorporated Project Implementation Parameters in the annual MOU** being signed between my Ministry & CPSUs and the system of Quarterly Performance Review of the CPSUs.

Q | Minister, could you elaborate on government's plans of establishment of a carbon market?

We already have one of the largest energy saving schemes in the world. The new carbon market will further enhance the energy transition efforts and will help the country to achieve the NDC goal of reducing Emissions Intensity of the GDP by 45 percent by 2030 against 2005 levels.

GHG emissions intensity benchmark and targets will be developed for the sectors, which will be aligned with India's emissions trajectory as per climate goals. The carbon credits will be earned based on the performance against these sectoral trajectories.

The key aspect of the scheme is allowing the participation of non-obligated entities who also can also register de-carbonization projects and generate carbon credits. Validation of carbon credits will be done. India is a big market and can very well have its own standards and systems commensurate with international standards.

Q | Where have we reached in terms of government's commitment to transitioning to non-fossil fuel sources - renewable energy in view of the COP-21 targets globally and what are the future plans?

We are the leaders in the fight against climate change. Our NDC target of 40% installed capacity from non-fossil fuel sources was achieved 9 years in advance. We have 190 GW of installed capacity from non-fossil fuel sources today. This is 44% of the total installed capacity. We have now committed a target of 50% of non-fossil capacity by 2030 but with the current policies laid out we are hopeful to even reach a level of 65% by 2030.

As much as 100GW of wind, solar, hydro and pumped storage is under implementation as we speak. Our Transmission plan to evacuate this power is also in place. The pace of RE capacity addition in India is fastest among all the major economies in the world as per a recent report by IEA.

We have 28,400 MW of thermal capacity under construction, 9000 MW under bidding and about 50,000 MW under various clearances.

Q | What are some of the strategies implemented by the government to achieve substantial emissions reduction and enhance energy efficiency, establishing our nation as a global climate action leader?

We surpassed our goal of reducing emissions intensity by 33% in 2019, which was 11 years ahead of schedule.

We have energy efficiencies measures like Building Energy Efficiency, Star Label for Consumer Goods, Energy Efficiency in Small and Medium Industry, Ujala, Street Light National Programme etc.

This has avoided 306 million tonnes of CO2 emission annually.

Total energy savings per year have sky rocketed from 3.02 million tonnes of Oil equivalent (toe) to a staggering 50.8 million toe.

We have revised our NDCs to 45% cut in GDP emissions by 2030.

Q | Going forward, what strategic vision does the government hold for the future development of India's Power Sector?

Although we have come a long way in the last decade, but the journey is ongoing. We cannot afford to take our foot off the pedal. We will continue with steps to make the system modern and viable.

The first challenge is capacity. Our economy is growing rapidly – it can continue growing rapidly only if there is electricity. We have to increase our generation capacity rapidly.

In the coming years, AT&C Losses will be brought down to 12%.

We need viable Grid Scale Storage, Green Hydrogen and Green Ammonia along with Pumped Storage to align with the upcoming Renewable Capacity. We have come out with a policy for pumped storage; and PLI schemes for Green Hydrogen and Battery Storage.

The sector has attracted huge investments and will continue to do so.

We think for not only the present but future generations too. This is New India, we are ready to claim our rightful place in the world. We have the vision, the will, the capability and the commitment to make India a developed country by 2047.